

TOMAS ESPAÑA

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EXPERIENCE

Scuola Normale Superiore of Pisa

Pisa, Italy

Research Intern

4 months – 2025

- Reinforcement Learning for Optimal Execution with the Queue Reactive Model
- Supervisors: Yadh Hafsi, Edoardo Vittori, Fabrizio Lillo

EconophysiX - CFM Chair of Econophysics & Complex Systems at Ecole Polytechnique

Paris, France

Research Intern

6 months – 2024

- High-dimensional Statistics, Random Matrix Theory, Portfolio Optimization
- Supervisors: Victor Le Coz, Matteo Smerlak, Jean-Philippe Bouchaud

CFM - Capital Fund Management

Paris, France

Quantitative Research Intern

6 months – 2023

- Market Latency (Queueing Theory & Optimization) and Market Microstructure for the Execution Strategies team
- Coded a latency simulator of the CFM order pipeline to improve execution speed.

CentraleSupélec - Université Paris-Saclay

Paris, France

Math Teaching Assistant

5 months – 2023

- Taught Measure Theory and Partial Differential Equations to last year BS students.

EDUCATION

Operations Research & Financial Engineering - Princeton University

Princeton, USA

First-year PhD student

2025 – Present

M2MO - Université Paris Cité

Paris, France

MS in Mathematics: Probability, Finance & Data Science

2024 – 2025

- One of the top-ranking MS in Probability and Finance (ex DEA Laure Elie). Double degree with CentraleSupélec.

CentraleSupélec - Université Paris-Saclay

Paris, France

BS and MS of Science and Engineering - French Engineering "Grande Ecole"

2021 – 2025

- Applied Mathematics

PUBLICATIONS

Journal Articles:

- P. Bousseyroux, T. Espana, M. Smerlak. *Another Marčenko-Pastur for Kendall's Tau*. Electronic Communications in Probability (ECP), 2025

Pre-prints:

- T. Espana, Y. Hafsi, F. Lillo E. Vittori. *Reinforcement Learning in Queue-Reactive Models: Application to Optimal Execution*. arXiv preprint, 2025
- F. Benaych-Georges, T. Espana. *Spectral Properties of Generalized Correlation Coefficients*. arXiv preprint, 2025
- T. Espana, V. Le Coz, M. Smerlak. *Kendall Correlation Coefficients for Portfolio Optimization*. arXiv preprint, 2024

CONFERENCES

- SIAM Conference on Financial Mathematics and Engineering, Miami, July 2025.
- Poster Presentation. 18th Financial Risks International Forum, Paris, March 2025.
- Poster Presentation. Brunel-Bielefeld Workshop on Random Matrix Theory and Applications, London, December 2024.
- Poster Presentation. Random Tensors and Related Topics, Paris, October 2024.